Ex No: 7 FLOW CONTROL AT DATA LINK LAYER

Date: 24/09/26

AIM:

write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another

Sender.py

```
import time
import os

def save_data(file_name, data):
    with open(file_name, "w") as file:
        file.write(data)

def get_data(file_name):
    if not os.path.exists(file_name):
        return None
    if os.path.getsize(file_name) == 0:
        return None
    with open(file_name, "r") as file:
        return file.read().strip()

def empty_file(file_name):
    if os.path.exists(file_name):
        open(file_name, "w").close()

def sender_program():
    Print("=======")
    Print("SENDER PROGRAM")
    Print("=======")

size = int(input("Enter window size: "))
text = input("Enter Text Message: ")
length = len(text)
start = 0
```

```
empty_fb("Sender_Buffer.txt")
empty_fb("Receiver_Buffer.txt")
while start < length:
    end = min(start + Size, length)
    frames = ""
    Print("\n----------")
    Print("Sending Frames")
    Print("-----")
    for position in range(start, end):
        frame_number = position
        character = text[position]
        Print(
            "Frame No:", frame_number,
            "DATA:", character)
        frames += f"{frame_number}: {character}\n"
    save_data(
        "Sender_Buffer.txt", frames.strip())
    Print("\n Sender_Buffer.txt updated.")
    Print("\n Waiting 10 Seconds...")
    Print("You Can Manually change a frame number"
          "in Sender_Buffer.txt.")
    time.sleep(10)
    reply = None
    for attempt in range(5):
        reply = get_data("Receiver_Buffer.txt")
        if reply:
            break
        time.sleep(1)
        if reply:
            Print(
                "\n Receiver Response:", reply
                )
```

```
try:
    response_type, response_number = reply.split(":")
    response_number = int(response_number)

    if response_type == "ACK":
        print("ACK received for frame", response_number)
        print("Sliding Window is moving forward...")
        start = response_number + 1
    elif response_type == "NACK":
        print("Nack received for Frame", response_number)
        print("Retransmitting from Frame", response_number)
        start = response_number
    else:
        print(
            "Invalid response received.")
except ValueError:
    print("Invalid acknowledgement format.")

else:
    print("\n Timeout!")
    print("No acknowledgement received.")
    print("Current window will be retransmitted.")

empty_file("Receiver_Buffer.txt")

print("\n =====")
print("Transmission Completed!")
print("All Frames were acknowledged.")
print("======")
save_data("Sender_Buffer.txt", "Done")

if __name__ == "__main__":
    Sender program()
```

Receiver.py

```
import time
import os
def save_response(file_name, data):
    with open(file_name, "w") as file:
        file.write(data)
def get_response(file_name):
    if not os.path.exists(file_name):
        return None
    if os.path.getsize(file_name) == 0:
        return None
    with open(file_name, "r") as file:
        return file.read().strip()
def clear_buffer(file_name):
    if os.path.exists(file_name):
        open(file_name, "w").close()
def receiver_program():
    print("======")
    print("RECEIVER PROGRAM")
    print("======")
    next_frame = 0
    message_data = []
    print("\nReceiver is waiting for frames...")
    while True:
        time.sleep(1)
        incoming_data = get_response("Sender_Buffer.txt")
        if not incoming_data:
            continue
        if incoming_data == "DONE":
            print("\nTransmission completed by Sender.")
            print("Final Message:", "".join(message_data))
```

```
print("\n-------")
print("Reading Sender_Buffer.txt")
print("-------")
frame_list = incoming_data.split("\n")
error = False
for frame_data in frame_list:
    if frame_data == "":
        continue
    try:
        number, value = frame_data.split(":")
        number = int(number)
        if number == next_frame and not error:
            print("Frame", number, "Data:", value, "→CORRECT")
            message_data.append(value)
            next_frame += 1
        else:
            print("Frame", number, "Data:", value, "→ERROR")
            print("Expected Frame:", next_frame)
            Error = True
    except ValueError:
        print("Invalid frame format.")
        Error = True

if error:
    print("\n Sending Nack:", next_frame)
    save_response("Raiver_Buffer.txt", f"NACK:{next_fra
else:
    acknowledged_frame = next_frame - 1
    if acknowledged_frame >= 0:
        print("\n Sending ACK:", acknowledged frame)
        "Raiver_Buffer.txt", f"ACK:{acknowledged
```

```
clear_buffer("Sender_Buffer.txt")
if __name__ == "__main__":
    receiver-program
```

OUTPUT:

```
= = = = = = = = = = = = =
      SENDER PROGRAM
= = = = = = = = = = = = =
Enter Window Size : 3
Enter Text Message : hello

- - - - - - - -
Sending Frames
- - - - - - - - -
Frame No : 0        DATA : h
Frame No : 1        DATA : e
Frame No : 2        DATA : l
Sender_Buffer.txt updated
Waiting 10 sec...
You can manually change a frame number in Sender_Buffer.txt

= = = = = = = = = = = = =
      RECEIVER PROGRAM
= = = = = = = = = = = = =
Receiver is waiting for frames...
- - - - - - - - - - - -
Reading Sender_Buffer.txt
- - - - - - - - - - - -
Frame 0  DATA : h -> Correct
Frame 1  DATA : e -> Correct
Frame 2  DATA : l -> Correct
Sending ACK : 2
Received Message : hel
```

RESULT:

Thus the sliding window protocol was successfully implemented using Python.